Type MS · Explain systolic, diastolic and changing pressure around the circulation

Answer 1 of 8 · Question 3 part b · 5 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

Number				
3	(a) (i)	• 120 - 5; • = 115 mm Hg;	allow +/- 2 for each figure ECF	2
	(ii)	A; (arteries) B/C/D are represented by other parts of the graph		1
	(iii)	D; (veins) A/B/C are represented by other parts of the graph		1
	(b)	any 5 from • heart; • (left) ventricle contracts; • forces blood out of heart/through aorta; • high pressure; • ventricle relaxes; • pressure falls;		5
	(c)	• capillaries; • no pulse/fluctuations; • as no muscle/elastic tissue; • (blood only pushed along at) low pressure/pressure drops;		4

Total 13 marks

Type MS · Explain systolic, diastolic and changing pressure around the circulation

Answer 1 of 8 · Question 3 part b · 5 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain systolic, diastolic and changing pressure around the circulation

Answer 2 of 8 · Question 3 part a.ii · 1 mark

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

Question number				
3	(a) (i)	• 120 - 5; • = 115 mm Hg;	allow +/- 2 for each figure ECF	2
	(ii)	A; (arteries) B/C/D are represented by other parts of the graph		1
	(iii)	D; (veins) A/B/C are represented by other parts of the graph		1
	(b)	any 5 from • heart; • (left) ventricle contracts; • forces blood out of heart/through aorta; • high pressure; • ventricle relaxes; • pressure falls;		5
	(c)	• capillaries; • no pulse/fluctuations; • as no muscle/elastic tissue; • (blood only pushed along at) low pressure/pressure drops;		4

Total 13 marks

Type MS · Explain systolic, diastolic and changing pressure around the circulation

Answer 2 of 8 · Question 3 part a.ii · 1 mark

2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain systolic, diastolic and changing pressure around the circulation

Answer 3 of 8 · Question 3 part a.iii · 1 mark

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.iii in this mark scheme extract

Question number				
3	(a) (i)	• 120 - 5; • = 115 mm Hg;	allow +/- 2 for each figure ECF	2
	(ii)	A; (arteries) B/C/D are represented by other parts of the graph		1
	(iii)	D; (veins) A/B/C are represented by other parts of the graph		1
	(b)	any 5 from • heart; • (left) ventricle contracts; • forces blood out of heart/through aorta; • high pressure; • ventricle relaxes; • pressure falls;		5
	(c)	• capillaries; • no pulse/fluctuations; • as no muscle/elastic tissue; • (blood only pushed along at) low pressure/pressure drops;		4

Total 13 marks

Type MS · Explain systolic, diastolic and changing pressure around the circulation

Answer 3 of 8 · Question 3 part a.iii · 1 mark

2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain systolic, diastolic and changing pressure around the circulation

Answer 4 of 8 · Question 3 part c · 4 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

number				
3	(a) (i)	• 120 - 5; • = 115 mm Hg;	allow +/- 2 for each figure ECF	2
	(ii)	A; (arteries) B/C/D are represented by other parts of the graph		1
	(iii)	D; (veins) A/B/C are represented by other parts of the graph		1
	(b)	any 5 from • heart; • (left) ventricle contracts; • forces blood out of heart/through aorta; • high pressure; • ventricle relaxes; • pressure falls;		5
	(c)	• capillaries; • no pulse/fluctuations; • as no muscle/elastic tissue; • (blood only pushed along at) low pressure/pressure drops;		4

Total 13 marks

Type MS · Explain systolic, diastolic and changing pressure around the circulation

Answer 4 of 8 · Question 3 part c · 4 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain systolic, diastolic and changing pressure around the circulation

Answer 5 of 8 · Question 6 part b · 4 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

6 (a)	(i)	• 1 = artery/ aorta/arterioles/named artery; • 2 = vein/ venules/vena cava; • 3 = capillary; • 4 = vein;		4
	(ii)	• 1 = thick wall/muscular wall/elastic wall/fixed/rigid shape/small lumen; • 2 = thin wall/little muscle/little elastic tissue/ irregular shape/large lumen; • 3 = one cell thick/pores/gaps; • 4 = valves;		4
(b)		• M = (left) ventricle/heart; • N = arteries/arterioles/aorta/named artery; • O = capillaries; • P = veins/venules/vena cava;		4

Total 12 marks

Type MS · Explain systolic, diastolic and changing pressure around the circulation

Answer 5 of 8 · Question 6 part b · 4 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain systolic, diastolic and changing pressure around the circulation

Answer 6 of 8 · Question 5 part a.i · 4 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

5 (a)	(i)	• flows from artery; • where blood is at high pressure; • into vein; • because blood is at a low pressure;	mp1- accept in context of artery leaving heart mp3- accept in context vein of returning to the heart If comparative answer given for pressure score marking point 2 and 4	4
	(ii)	glomerulus;		1
	(iii)	• large (surface) area; • for diffusion / molecules/ named molecules to pass out(of blood);	More area for diffusion = 2 marks	2
	(iv)	more urea;		1
(b)		• no/less gradient between liquid X and blood (glucose)/blood and X have same concentration/eq; • glucose does not pass out/ glucose is not lost: • by diffusion • needed for respiration; • minimises osmotic effect;		3
(c)		• protein digested/broken down into <u>amino acids</u>; • (excess) **<u>amino acids</u>** broken down/deaminated; • urea; • less protein means less urea produced; • less urea means less to be filtered/removed ; • high concentration of urea/nitrogenous waste can be toxic;	MP3– accept urea from protein breakdown	4
				Total =15

Type MS · Explain systolic, diastolic and changing pressure around the circulation

Answer 6 of 8 · Question 5 part a.i · 4 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain systolic, diastolic and changing pressure around the circulation

Answer 7 of 8 · Question 9 part b.i · 2 marks

2022 JUN · Paper 01 · find part b.i in this mark scheme extract

9 (a)	• beta blockers; • block/bind to receptors; • which prevents the release of adrenaline/noradrenaline; • heart rate decreases; OR • ACE inhibitors; • reduce the activity of enzymes; • less angiotensin; • less narrowing of blood vessels;		4
(b)(i)	any two from • how much pressure blood is exerting; • against artery walls; • when ventricles contract/pump;		2
(ii)	• pressure in the arteries; • when the heart/ventricles at rest/relaxed/between beats;		2
			Total 8

Type MS · Explain systolic, diastolic and changing pressure around the circulation

Answer 8 of 8 · Question 9 part b.ii · 2 marks

2022 JUN · Paper 01 · find part b.ii in this mark scheme extract

9 (a)	• beta blockers; • block/bind to receptors; • which prevents the release of adrenaline/noradrenaline; • heart rate decreases; OR • ACE inhibitors; • reduce the activity of enzymes; • less angiotensin; • less narrowing of blood vessels;		4
(b)(i)	any two from • how much pressure blood is exerting; • against artery walls; • when ventricles contract/pump;		2
(ii)	• pressure in the arteries; • when the heart/ventricles at rest/relaxed/between beats;		2
			Total 8